

DEXTER FERNANDES

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github.com/Dexter-Fernandes

SUMMARY

Computer Vision Engineer with 3+ years building and deploying real-time vision systems on edge hardware. Owned production pipelines end to end on NVIDIA DeepStream and OpenVINO for BP drilling rigs and Austin Industries construction sites, from data curation and training through quantisation, C++ deployment and on-site validation. MSc Robotics (Merit), University of Bristol.

RELEVANT EXPERIENCE

Esbaar — AI Engineer

01/2023 – 08/2023 · MUSCAT, OMAN

- Owned the end-to-end ML lifecycle for real-time perception on BP drilling rigs, from data curation and training through deployment and on-site validation, improving F1 by 15% under 24/7 operation.
- Developed and maintained NVIDIA DeepStream 6.x and FFmpeg video pipelines for 24/7 perception in desert conditions, covering camera ingestion, inference and field diagnostics.
- Hardened deployed services with Dockerised microservices, Kafka messaging and a watchdog exposing health and restart control over REST, cutting live video pipeline crashes by 67%.
- Adapted YOLOv5 detectors in PyTorch for rig-specific classes with Optuna sweeps and Weights & Biases tracking, benchmarking against EfficientDet on accuracy, latency and stability.
- Applied image preprocessing and artefact detection with OpenCV and scikit-image to catch mud and sand contamination on camera feeds, reducing false positives by 13%.
- Built ingestion and validation pipelines with Airflow, MLflow, DVC and CVAT, and added pytest and linting to CI, cutting rollout from 72h to under 24h.

LivNSense Technologies — Lead Software Engineer, AI

05/2020 – 12/2022 · BENGALURU, INDIA

- Led four engineers building a two-stage detector feeding helmet-colour classification for construction safety in Texas, cutting false positives by 33% within latency and power budgets.
- Built asynchronous C++17 perception pipelines on OpenVINO across Intel Movidius VPUs and Core i7 industrial edge boxes, raising throughput from 5 to 15 FPS under real-time constraints.
- Optimised architectures with quantisation-aware training, structured pruning and layer fusion in Keras and OpenVINO, cutting model size by 50% while holding production accuracy.
- Built a real-time multi-camera perception pipeline on a moving construction vehicle, adding ByteTrack tracking to hold worker and vehicle identities and cut duplicate incident alerts.
- Deployed containerised ML inference microservices on Azure with REST endpoints and queue-based async processing, owning autoscaling, monitoring and cost-per-inference.
- Automated labelling by repurposing existing detection models and added dataset versioning with DVC, cutting manual labelling effort by 80% on large real-world datasets.

PROJECTS

DeepStream RTSP Pipeline github.com/Dexter-Fernandes/deepstream-rtsp-pipeline

06/2026

- Developed an end-to-end DeepStream 9.0 and GStreamer RTSP pipeline running YOLO detection and NvDCF tracking across 3 concurrent feeds, with metadata extraction, anonymisation and re-streamed output.
- Converted YOLO26n to a dynamic-batch TensorRT FP16 engine on 6GB VRAM Turing hardware, validating FP16 against FP32 accuracy and profiling p99 latency to sustain 29.7 FPS per stream.
- Implemented a C++ and CUDA TensorRT decode plugin with IPluginV2DynamicExt, moving post-processing into the inference engine to support custom-layer deployment.
- Benchmarked IoU, NvDCF and NvSORT trackers on MOTA, IDF1, ID switches, fragmentation, FPS and VRAM impact to guide tracker selection.

Modern Edge CV Benchmarking github.com/Dexter-Fernandes/ModernEdge-CV-Benchmarking

03/2026

- Benchmarked modern PyTorch detectors including YOLOv8, YOLOv10, YOLO26 and RT-DETR on accuracy, latency and model size to compare deployment suitability for edge targets.
- Extended the comparison to EfficientViT, MobileSAM and Depth Anything v2 to assess whether segmentation and depth models are viable under the same edge constraints.
- Evaluated Vision Transformer and foundation-model approaches across detection, segmentation, depth estimation and open-vocabulary understanding.
- Exported models to ONNX and tested OpenVINO deployment paths, comparing mAP, inference time and model size to support model selection for low-latency systems.

- Developed ROS2 nodes, launch files and parameterised configs in C++ and Python, using tf2, message synchronisation and timestamp alignment to debug transform timing issues.
- Evaluated LiDAR-centric and LiDAR plus RGB fusion SLAM in ROS2 Jazzy and Gazebo Harmonic, comparing GLIM and Cartographer across 2D and 3D LiDAR configurations.
- Implemented point cloud filtering, segmentation and registration with PCL and Open3D, added EKF localisation with robot_localization, and containerised the stack with Docker.

EDUCATION

University of Bristol — MSc Robotics, Merit

09/2023 – 09/2024 · BRISTOL, ENGLAND

Dissertation: Radiological SLAM with LiDAR Odometry

- Built a real-time SLAM system in C++17 and ROS2 (rclcpp) on a Clearpath Husky, fusing Velodyne VLP-16 LiDAR, RealSense D435i depth and radiological sensing for inspection mapping.
- Ran the full SLAM and perception stack on an NVIDIA Jetson AGX Xavier, working within the platform's compute and memory limits for on-robot operation.
- Calibrated LiDAR extrinsics to the robot base and resolved transform timing with tf2 across a PCL point cloud pipeline, benchmarking RTAB-Map, LIO-SAM and Cartographer on rosbag2 sequences.
- Configured Ceres, g2o and GTSAM as solver backends, tuned ICP parameters across environments, and set up Nav2 in simulation for waypoint traversal over the generated map.

Visvesvaraya National Institute of Technology — BTech Electrical and Electronics Engineering

07/2016 – 05/2020 · NAGPUR, INDIA

- Foundation in signals, systems, embedded hardware and control, which later supported work in real-time AI, sensor systems and robotics software.

SKILLS

Languages	Python, C++17, CUDA C++, Bash, SQL
Computer vision	Object detection, tracking, segmentation, defect and anomaly detection, optical flow, convolutional neural networks (CNNs), YOLO (v5/v8/v10/v26), RT-DETR, EfficientDet, PyTorch, TensorFlow, Keras, OpenCV, scikit-image, torchvision, NumPy
Edge inference	TensorRT (custom plugins, dynamic batching), OpenVINO, ONNX, quantisation-aware training, post-training quantisation, FP16/INT8, structured pruning, layer fusion, latency and VRAM profiling, Intel VTune, Intel Movidius VPU, NVIDIA Jetson AGX Xavier
Video pipelines	NVIDIA DeepStream 6.x/9.0, GStreamer, FFmpeg, RTSP, nvinfer, nvstreammux, nvtracker, NvDCF, NvSORT, ByteTrack, multi-camera synchronisation, ISP and camera tuning (RGB and IR)
Robotics and SLAM	ROS2 (Robot Operating System 2), rclcpp, rclpy, tf2, rosbag2, Gazebo, RTAB-Map, LIO-SAM, Cartographer, Nav2, LiDAR SLAM, ICP, pose graph optimisation, LiDAR extrinsics calibration, PCL, Open3D, Eigen, EKF
MLOps	MLflow, DVC, Apache Airflow, Apache Kafka, Optuna, Weights & Biases, CVAT, automated labelling, dataset versioning, synthetic data generation, benchmarking harnesses, ablation studies, Grad-CAM, TIDE, COCO annotation format
Production and cloud	Docker, CI/CD, Git, CMake, pytest, unittest, gtest, Microsoft Azure (Container Apps, Blob Storage, queues), REST API design, containerised microservices, multiprocessing, autoscaling, cost-per-inference, PostgreSQL, structured logging, monitoring, Linux
Classical ML	MultiRocket, gradient boosting, random forests, scikit-learn, Pandas, multivariate time-series classification
Professional	Technical leadership, mentoring, Agile and Scrum, Jira, stakeholder communication, cross-functional delivery

ADDITIONAL EXPERIENCE

GrowthStage — Technical Assessor, Computer Vision and Software Engineering

05/2026 – PRESENT · REMOTE

Assesses candidate technical depth across perception, SLAM and deployment using hands-on robotics and computer vision knowledge.

CERTIFICATIONS AND AWARDS

Deep Learning Specialisation, deeplearning.ai (07/2019) Best Speaker award, ABC Toastmasters Club (01/2020)